

Project Testing Phase

The **Project Testing Phase** is an important stage in the development of **EduGenie – Google Gemini Powered Learning Assistant**. In this phase, the completed system is tested to identify errors, check functionality, and ensure that the application works according to the requirements.

1. Purpose of Testing

The main purpose of testing is to:

- Identify and fix errors in the application.
- Check whether all features work correctly.
- Verify the AI response functionality.
- Ensure that the application is easy to use.
- Improve the reliability and performance of the system.

2. Types of Testing

1. Unit Testing

Each module is tested separately, such as the login module, question input module, and response module.

2. Integration Testing

Different modules are tested together to ensure proper communication between the frontend, backend, database, and Gemini API.

3. Functional Testing

The main functions of the application are checked to ensure that they work according to the requirements.

4. User Interface Testing

The interface is tested for readability, navigation, buttons, input fields, and overall ease of use.

5. API Testing

The connection with the Google Gemini API is tested to check whether questions are sent correctly and responses are received properly.

6. Performance Testing

The application is checked for response speed and stability when handling different questions.

3. Test Cases

Some sample test cases include:

Test Case	Expected Result
Enter a valid question	AI provides a relevant response
Submit an empty question	System displays an appropriate message
Ask a follow-up question	System responds appropriately
API connection available	Response is generated successfully
API connection unavailable	Suitable error message is displayed

Navigate between pages	Pages open correctly
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4. Error Handling

During testing, errors such as invalid input, API connection problems, server errors, and unexpected responses are identified. Proper error messages and handling mechanisms are added to improve the user experience.

5. Final Testing

After fixing the identified errors, the complete application is tested again. This ensures that changes made during development have not affected other features.

6. Expected Outcome

The Project Testing Phase ensures that **EduGenie is functional, reliable, user-friendly, and ready for implementation**. Successful testing confirms that the major system components work together as planned and that students can use the application to ask questions and receive AI-generated assistance.